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Paleomagnetism of the Catoctin volcanic province: A new Vendian-Cambrian apparent polar wander path for North America

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The Vendian/Cambrian segment of the Lauretian apparent polar wander path (APWP) has been poorly constrained and the subject of some controversy. The Catoctin volcanic province in central Virginia is well-dated at 570± 35 Ma (Rb-Sr) and 597 ± 18 Ma (U-Pb) and therefore presented an excellent paleomagnetic target for resolving the Laurentian Vendian-Cambrian APWP. A total of 206 samples from 32 sites were collected from the Catoctin basalts, feeder dikes and sills. The study revealed three ancient directions of magnetization. The youngest, C component, fails the fold test and yielded a characteristic in situ direction of D=147°, I= +44° (k=21, α95=9°). The corresponding paleopole falls along the Middle Ordovician segment of the Laurentian APWP and we consider this component to be the result of a Taconic remagnetization. The second component, the B component, is carried by hematite, exhibits dual-polarities and passes a fold test. The tilt-corrected B component characteristic direction is D=92°, I= +17° (k=16, α95=13°). The corresponding paleopole at 4°S, 193°E falls near a well-established Late Cambrian (505 Ma) pole for Laurentia, and we consider this component to be a remagnetization during a Late Cambrian tectonic event in the central Appalachians. The third component isolated in the Catoctin basalts, the A component, yields a tilt-corrected mean of D=68°, I= +84° (k=59, α95=9°). This component passes a fold and reversal test. A suite of samples was collected from two Catoctin feeder dikes and surrounding country rocks that yield a positive baked contact test. The A pole at 43°S, 128°E falls significantly away from previously proposed Vendian poles for Laurentia. A reevaluation of previous paleomagnetic studies from coeval rock units reveals similarly steep directions and leads us to propose a new APWP. This new APW track indicates that Laurentia was located near the pole during the interval 615-580 Ma and drifted rapidly (16 cm yr^{-1}) toward its Late Cambrian equatorial position.

Introduction

The Wilson cycle of continental assembly and dispersal [*Wilson*, 1966] is one of the fundamental tenets of plate tectonic theory. The assembly of Pangea in the late Paleozoic and its subsequent Mesozoic dispersal is the best example of global scale, Wilson cycle, tectonism. It is postulated that an earlier aggregation of continental landmasses occurred during the late Proterozoic [*Dalziel*, 1992; *Hoffman*, 1991; *Piper*, 1987; *Bond et al.*, 1984; *Piper*, 1982]. The assembly and dispersal of this supercontinent, recently called Rodinia [*McMenamin and McMenamin*, 1990], coincided with major changes in biological evolution, seawater chemistry and, by inference, paleoclimate. Most of the arguments favoring Rodinia were based on geologic evidence because the paleomagnetic evidence was considered of insufficient quality to provide rigorous tests of continental configurations [*Van der Voo and Meert*, 1991]. Nevertheless, paleomagnetism offers the only quantitative tool for determining pre-Mesozoic plate motions and plate configurations. Laurentia occupied the central part of Rodinia, and detailed knowledge of its paleogeographic history during the Vendian will provide the initial framework for the paleogeography of the surrounding landmasses.

The Vendian-Early Cambrian (650-540 Ma) portion of the Laurentian apparent polar wander

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path (APWP) is poorly constrained and the subject of recent controversy. The problem in interpreting this segment of the APWP arises from inconsistencies in placing the exact Vendian/Cambrian boundary in a radiometrically calibrated time scale and problems in interpreting disparate poles of similar age. New estimates for the age of the Vendian/Cambrian boundary range between 525 and 560 Ma with the most commonly cited figure of 540 Ma [*Link et al.*, 1992]. We acknowledge the controversy regarding the age of the Vendian/Cambrian boundary and accept the estimate of *Conway Morris* [1989] of 540 Ma as the base of the Cambrian. The paleogeographic controversy was highlighted by *Symons and Chiasson* [1991] after their study of the 577 Ma Callander Complex of Ontario, Canada. This well-dated pole falls at 46°S, 121°E, away from any previously proposed segment of the Laurentian path such as the APWP of *Van der Voo* [1990]. The Catoctin volcanic province in Virginia was dated at 573 ± 35 Ma using Rb-Sr by *Badger and Sinha* [1988], and a later study of interbedded rhyolites produced a U-Pb zircon age of 597 ± 18 Ma [*Aleinikoff et al.*, 1991]. Thus the Catoctin basalts presented an excellent target for further resolution of the Vendian/Cambrian APWP. We sampled Catoctin basalts, feeder dikes and sills at 32 sites along the Blue Ridge in central Virginia (Figure 1).

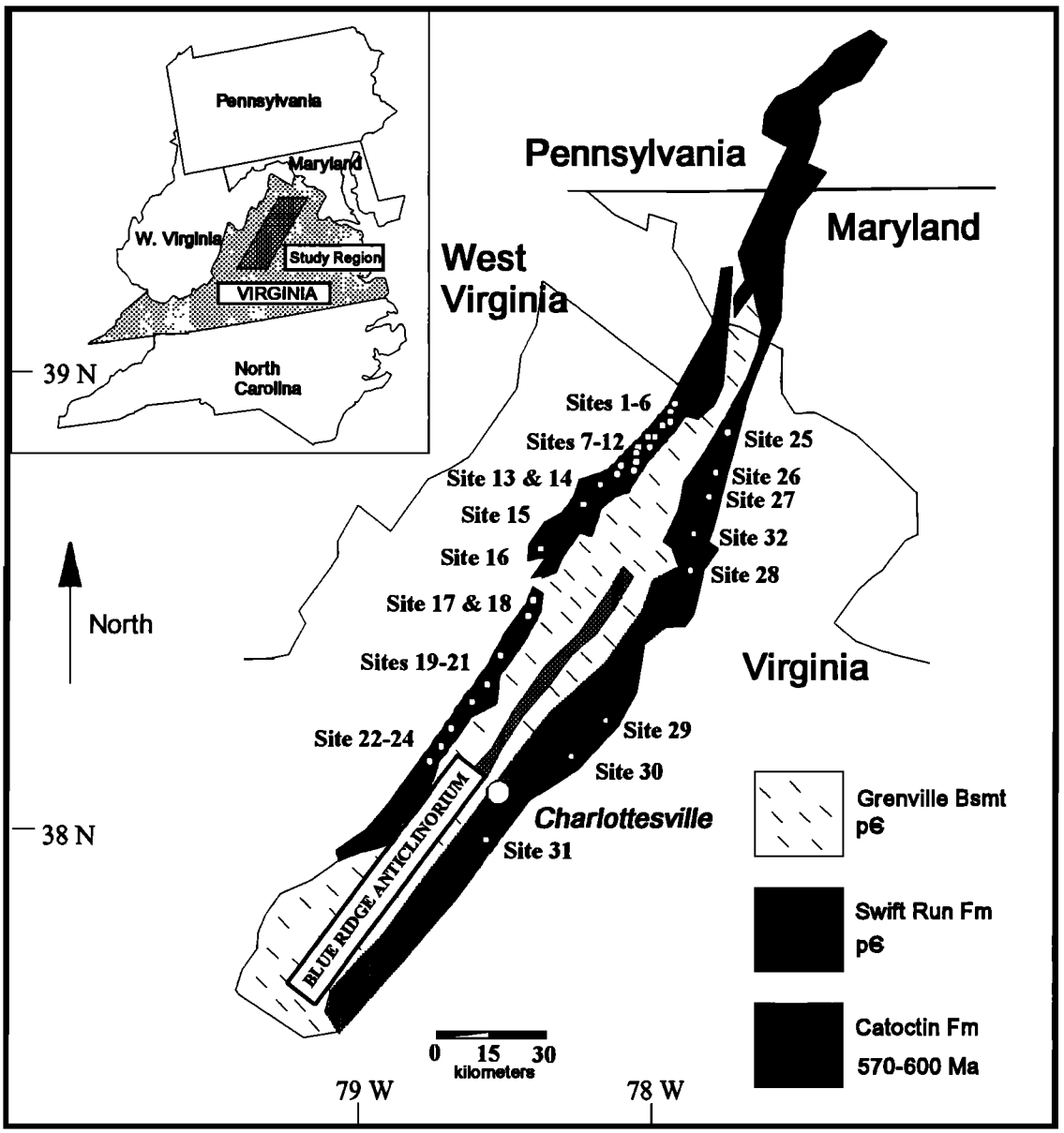


Figure 1. Map of the study area showing the location of paleomagnetic sampling sites along the Blue Ridge anticlinorium in Virginia. Most of the sampling was confined to an area within Shenandoah National Park. Sites 5,8,10 and 28 were Catoctin feeder dikes of sills.

GEOLOGY

The Catoctin volcanic province consists of predominately subaerial basalt flows that flank both limbs of the Blue Ridge anticlinorium (Figure 1). This major flood basalt province in the eastern United States is associated with the 600-550 Ma breakup of a late Proterozoic supercontinent and the opening of the Iapetus Ocean [*Bond et al.*, 1984; *Dalziel*, 1991]. The basalts vary in metamorphic grade from amphibolite facies in Maryland and Pennsylvania to lowest greenschist facies in northwestern Virginia and increase again to uppermost greenschist in southwestern Virginia [*Reed and Morgan*, 1971; *Badger and Sinha*, 1988]. The metamorphic grade also varies within individual massive flows. The central portion of any single flow is of lower metamorphic grade than the top or bottom of the flow [*Badger and Sinha*, 1988]. *Badger and Sinha* [1988] estimated the age of this metamorphism at 450 Ma (Rb-Sr) in agreement with the estimates provided by *Pettingil et al.* [1984]; *Reed and Morgan* [1971]; and *Cloos* [1968].

The Grenville basement rocks in the area (Pedlar Granite and Old Rag Granite) are cut by numerous diabase and metadiabase dikes. These dikes are thought to be feeders to the Catoctin lava flows based on their spatial relationship, metamorphic grade, petrographic similarities, absence in younger rocks, and bulk chemistry [*Reed and Morgan*, 1971; *Mose and Nagel*, 1984; *Reed and Clarke*, 1989]. These dikes range in thickness from a few centimeters to 20 m [*Reed and Morgan*, 1971; *Reed*, 1955].

TECTONIC HISTORY

Recent geologic evidence suggests that Laurentia formed the interior portion of a supercontinent that experienced two rifting events during latest Proterozoic time (Figure 2) [*Hoffman*, 1991; *Dalziel*, 1992]. The earlier rifting event (750-680 Ma) is associated with the separation of proto East Gondwana (Australia, Antarctica and India) from the present-day western margin of Laurentia. *Powell et al.* [1993] present paleomagnetic evidence for this rifting at 700 Ma. A synchronous, failed, rifting event occurred along the present-day eastern margin of Laurentia [*Bond et al.*, 1984; *Hoffman*, 1991]. There is considerable evidence of this failed rift event in the study area represented by the Mount. Rogers and Grandfather Mountain volcanic rocks of North Carolina and southern Virginia and by the Crossnore plutonic suite of Virginia. The Franklin dikes and sills of Canada are also associated with this failed eastern rift system. The Ice Brook and Varangian

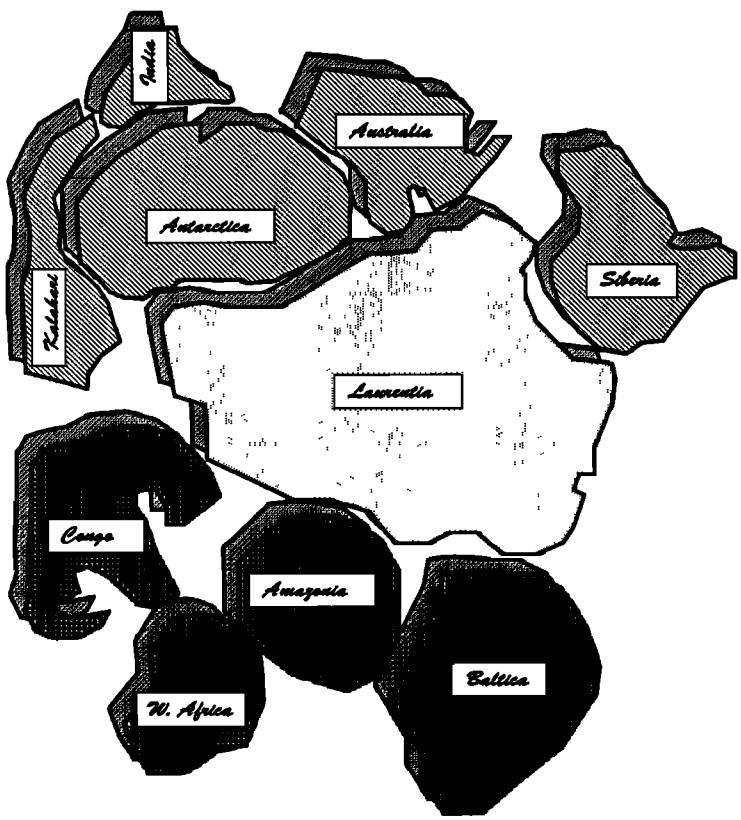


Figure 2. The Late Proterozoic supercontinent (ca. 750 Ma) according to *Hoffman* [1991]. Laurentia occupies the interior portion of the supercontinent that "breaks out" at 600 Ma following an earlier separation when East Gondwana rifted away from Laurentia.

glaciations on Laurentia fill the hiatus between rifting events. The second rifting event began at 615 Ma and continued until about 560 Ma [*Glover*, 1989]. Plutonic-volcanic events of this age are found along the eastern margin of North America. Figure 3 shows the units that were studied paleomagnetically and includes the 615 Ma Long Range dikes in Labrador at the northern end to the 560 Ma Unicoi basalts in Tennessee at the southern terminus. This second rifting event is manifested by the outpouring of the 570-600 Ma Catoctin volcanic rocks.

The Early Cambrian rift drift sequence includes the basal Chilhowee sequence of nonmarine feldspathic sandstones and conglomerates, the Unicoi basalts, the Middle Cambrian, red-clastic, Rome-Waynesboro Formations and finally the Middle Cambrian Shady and Tomstown dolomites [*Glover*, 1989; *Simpson and Eriksson*, 1989]. Subsequently, a pre-Taconic event in the Appalachians is well substantiated by *Glover* [1989] and is ascribed to a collision of the Carolinian (Piedmont) terrane between 510 and 490 Ma. *Pavlides* [1989] identified a Cambrian arc in the Piedmont of Virginia which he considered indicative of an early 505 Ma collisional event. A similar Late Cambrian event is called the Penobscottian in Maine [*Neuman*, 1967; *Boone and Boudette*, 1989].

The ancestral Blue Ridge was formed when this portion of the Laurentian margin was thrust westward during the early phases of the Taconic orogeny [*Glover*, 1989; *Wehr and Glover*, 1985]. Many of the Taconic structural features were overprinted during the later Alleghenian orogeny; nowever, as mentioned above, there is good isotopic age evidence for the Taconic event in the Blue Ridge. Deformation of the Catoctin volcanic province is variable. Dips of individual fold limbs range from 8° in the northern section of the study area to 125° in the overturned limbs near Charlottesville, Virginia. In general, deformation increases from north to south in the study region.

SAMPLING

A total of 206 samples were collected from 32 sites in Catoctin basalts, dikes and sills along a 200 km transect on both flanks of the Blue Ridge anticlinorium in western Virginia for paleomagnetic investigation (Figure 1). Of these 32 sites, four sites were sampled from Catoctin

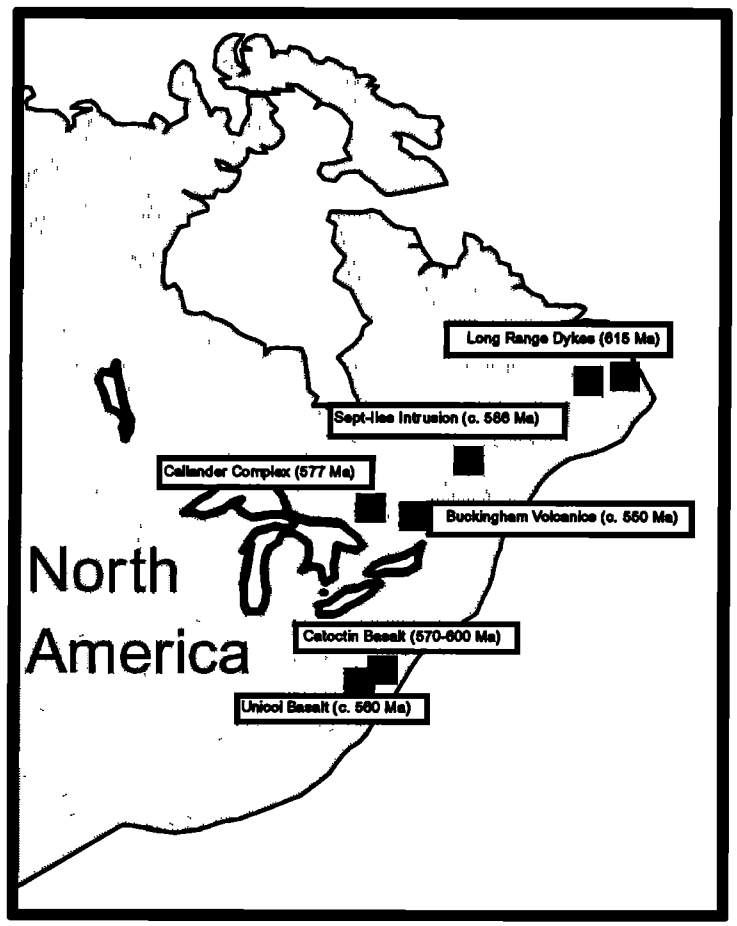


Figure 3. Map of Laurentia showing the locations of selected igneous events associated with the 615-550 Ma opening of the Iapetus Ocean.

feeder dikes, sills and country rock of the Pedlar granite for a baked contact test. The samples were drilled in the field except for two sites for which specimens were drilled in the laboratory from oriented hand samples. Drilled samples were oriented using magnetic compass and inclinometer. The sampling transect included samples of a range of metamorphic grades with higher-grade upper greenschist samples generally found near the southern terminus of Skyline Drive in Shenandoah National Park near Charlottesville, Virginia.

The intensities of the natural remanent magnetization (NRM) range between 10^{-5} and 10 A/m as measured on either a ScT cryogenic magnetometer or Schonstedt spinner magnetometer. Thermal and alternating field (AF) demagnetization carried out on a pilot selection of samples for each site were used to determine the optimum method of demagnetization. In general, both treatments yielded similar results; nevertheless, most samples were treated thermally because complete decay of the NRM could be achieved. Demagnetization results were plotted on orthogonal vector diagrams [*Zijderveld,* 1967], and linear trajectories were used to determine directions of magnetic components by a least squares fit comprising three measurements or more [*Kirschvink,* 1980]. Occasionally, linear components were not isolated because of overlapping coercivity or unblocking temperature spectra; for such results, great circle analysis was combined with stable endpoint data to determine the site means following the methods of *Bailey and Halls* [1984] and *Halls* [1978]. At least three distinct, ancient magnetizations were isolated in the samples and will be discussed sequentially below. A fourth and common, low-temperature component was found at 12 sites yielding a mean of D=15°, I=+61° (k= 74, α95= 7°) that we attribute to a viscous overprint of recent origin.

C Component

The results for sites containing the C component are given in Table 1. This component has a characteristic unblocking temperature between 400° and 580°C and is the most common component observed in the Catoctin samples with 84 samples from 15 sites showing this magnetization. Figure 4 shows typical orthogonal vector plots for these samples and stereographic plots of the in-situ and tilt-corrected site mean directions. Both the unblocking temperature spectra and the response of these rocks to isothermal remanence acquisition (IRM) indicate that the carrier of this magnetization is magnetite (Figure 5). Only a single polarity was observed in samples carrying this direction. The characteristic in-situ site mean direction of the C component is D=147°, I=+44° (k=21, α95=9°) and the tilt-corrected site mean direction is D=143°, I=+24° (k=3, α95=26°). A fold test conducted for the C component using the *McFadden* [1990] test demonstrates that the hypothesis that assumes the magnetization was acquired before tectonic deformation can be rejected at the 99% confidence level (ξ tilt-corrected=6.406 and ξ 99=6.305 for n=15). Samples that carried the C component were collected from basalts showing higher metamorphic grade and greater deformation, coming mostly from steeply dipping and overturned limbs with strong cleavage development.

B Component

This component was isolated in 50 samples from nine sites. The characteristic unblocking temperature for the B component was between 640° and 690°C. Figure 6 shows typical orthogonal vector plots for these samples and stereographic plots of the in situ and tilt-corrected site mean directions. The unblocking temperature spectra and the response of these samples to IRM acquisition (Figure 5) indicate that hematite is the carrier of the remanence in these samples. The site mean results for the B component are given in Table 1. The mean of the in situ directions for the B component is D=91°, I=+29° (k=8 α95=20°) and the tilt-corrected mean is D=92°, I=+17° (k=16 α95=13°). Both polarities were observed in the samples from these nine sites (Figure 6). The fold test conducted for the B component directions demonstrates that the hypothesis which assumes the magnetization was acquired in situ can be rejected at the 99% confidence level (ξ in situ=4.997 and ξ99=4.85 for n=9). This magnetization was therefore acquired before deformation.

A Component

The third magnetization component isolated in the Catoctin samples, the A component, was found in 30 samples from eight sites. These samples include eight Pedlar Granite samples from contact zones next to Catoctin feeder dikes. Table 2 gives the site mean results for the A component and Figure 7 shows typical orthogonal vector diagrams for these samples. The magnetic carrier of remanence A is magnetite as indicated by unblocking temperatures between 500° and 585°C and IRM studies (Figure 5). The A component at site 2 in a porphyritic basalt is carried by hematite. A contact test was conducted on samples from site 8, where a 4 m wide dike intruded the Pedlar Granite. Orthogonal vector plots are shown for the dike (Figures 8a and 8b);

TABLE 1. Paleomagnetic Results

	C Component							B Component										
Site	n/N	Ds	Is	k	α95	Plt	Plg	n/N	Ds	Is	Dc	Ic	k	α95	Plts	Plgs	Pltc	Plgc
1	3/6	150	+72	55	13	9 N	308 E	5/6	100	+43	102	+28	87	08	8 S	176 E	0 S	183 E
4	3/6	152	+34	24	25	27 N	140 E	6/6	103	+29	94	+31	13	19	1 S	182 E	7 S	186 E
6	4/6	111	+55	33	17	8 N	342 E											
10	3/5	150	+31	70	10	28 N	144 E											
11	8/8	164	+28	51	07	35 N	131 E											
12								5/5	070	+29	79	+18	65	12	26 S	202 E	15 S	202 E
13	4/8	131	+59	53	24	0 N	147 E											
16	3/5	137	+55	36	21	6 N	146 E											
17								4/5	107	-04	106	-16	20	25	14 S	013 E	17 S	018 E
18								6/6	251	-08	252	-06	100	08	18 S	211 E	16 S	211 E
19	4/5	157	+29	154	09	32 N	138 E											
20								4/5	102	-03	101	+16	20	22	11 S	015 E	03 S	009 E
21	3/3	146	+30	21	26	27 N	148 E											
22	17/17	143	+58	27	07	21 N	168 E											
23	8/8	155	+44	50	07	21 N	136 E											
24	6/6	159	+45	30	14	22 N	132 E											
25								7/7	51	+49	77	+20	27	10	48 S	198 E	17 S	202 E
26								5/5	117	+64	104	+35	26	15	12 S	152 E	2 S	178 E
29	6/6	129	+21	58	07	21 N	168 E											
30	6/8	144	+53	17	16	10 N	142 E	8/8	281	-30	279	-05	21	12	01 S	183 E	04 S	014 E
31	6/9	157	+37	24	14	27 N	136 E											
ISM	15 sites	147	+44	21	09	19 N	143 E	9 sites	91	+29			08	20	09 S	189 E		
TCM	15 sites	143	+24	03	26	28 N	154 E	9 sites			92	+17	16	13			04 S	193 E

The site means are calculated by giving each site unit weight. The paleomagnetic pole is calculated from the mean declination and inclination. n/N, number of samples used/number of samples per site; Ds, in situ declination; Is, in situ inclination; Dc, tilt-corrected declination; Ic, tilt-corrected inclination; k, Fisher precision parameter; α95, cone of 95% confidence; Plts, in situ virtual geomagnetic pole latitude; Plgs, in situ virtual geomagnetic pole longitude; Pltc, tilt-corrected virtual geomagnetic pole; latitude; Plgc, tilt-corrected virtual geomagnetic pole longitude; ISM, in-situ mean; TCM, tilt-corrected mean.

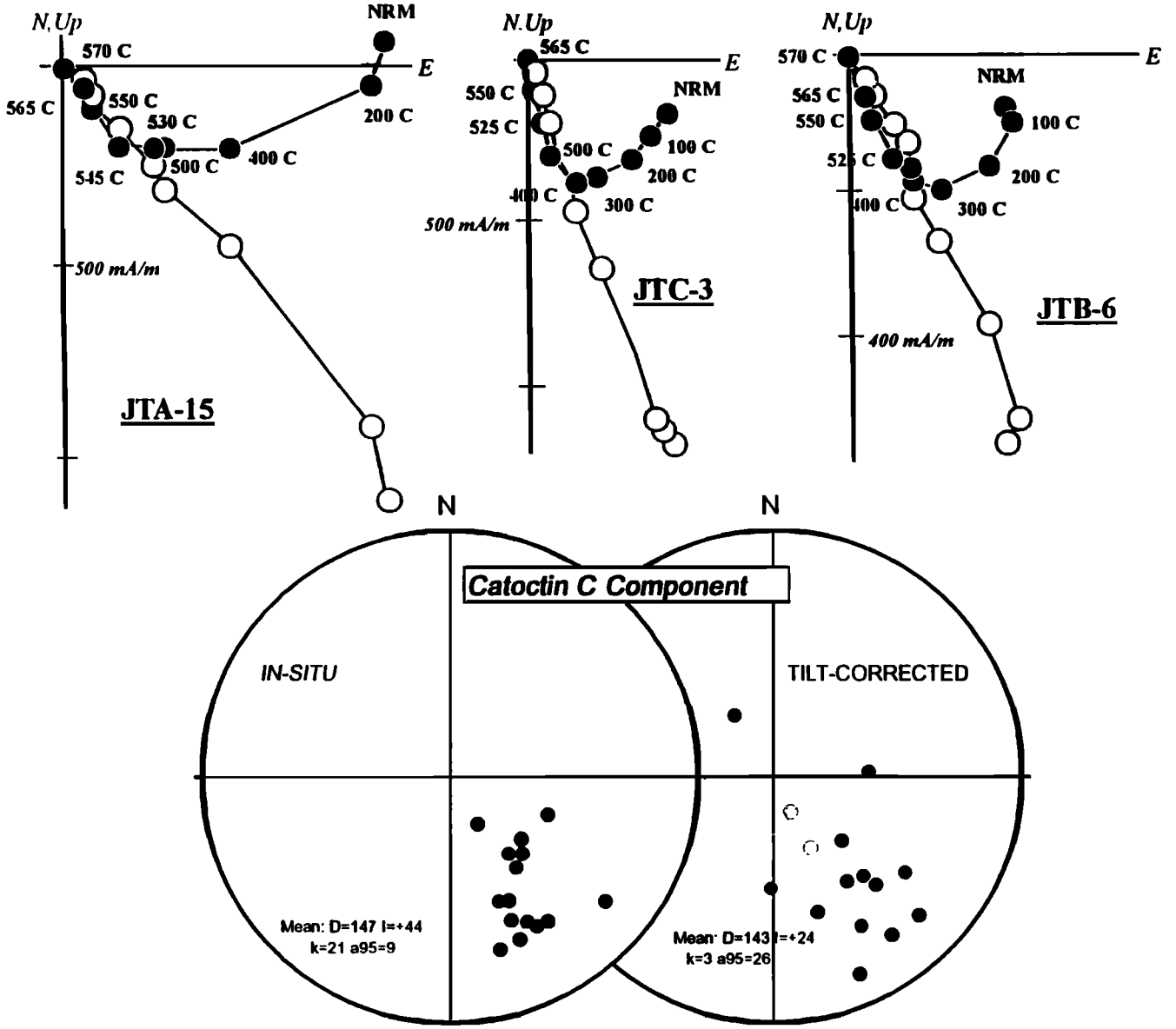


Figure 4. Typical in situ orthogonal vector diagrams of samples showing the C component of magnetization along with equal-area stereographic projections of the in situ and tilt-corrected site mean directions.

the granite 6 cm away from the dike contact (Figure 8c); the granite 2 m away from the dike (Figure 8d) and the granite 5 km from the dike (Figure 8e). The unbaked granite samples did not group well, but their inclinations were shallow-intermediate (less than 35 degrees). On the other hand, the directions of the dike and contact rocks are steep. Although false positive contact tests have been reported [*Jackson et al.*, 1988], the observed directions do not match any known younger directions for Laurentia, and these data are interpreted to suggest a positive contact test. A second positive contact test was obtained from site 28 where the Pedlar Granite was intruded by an eight meter wide sill.

The in situ mean direction for the A-Component sites is D=346°, I=+81° (k=23 α 95=14°) and the tilt-corrected site mean direction is D=68°, I=+84° (k=59 α95=9°). This magnetization shows dual polarities and is significantly different from both the B and C directions. The outlier fold test of *McFadden* [1990] for the site mean directions (Figure 7) is significant above the 95% confidence level (probability that the test statistic $\gamma m > 39° = .023$). If individual sample directions are used, the *McFadden* [1990] fold test indicates that the hypothesis which states the magnetization was acquired in situ can be rejected at the 92% confidence level (ξin situ=5.23 n=30). The results of these fold tests are interpreted to indicate that the A component magnetization is prefolding. The *McFadden and McElhinny* [1990] reversals test is inconclusive for the in situ directions but is positive for the tilt-corrected directions and is rated C class. We consider the evidence to favor a primary origin for the A component magnetization.

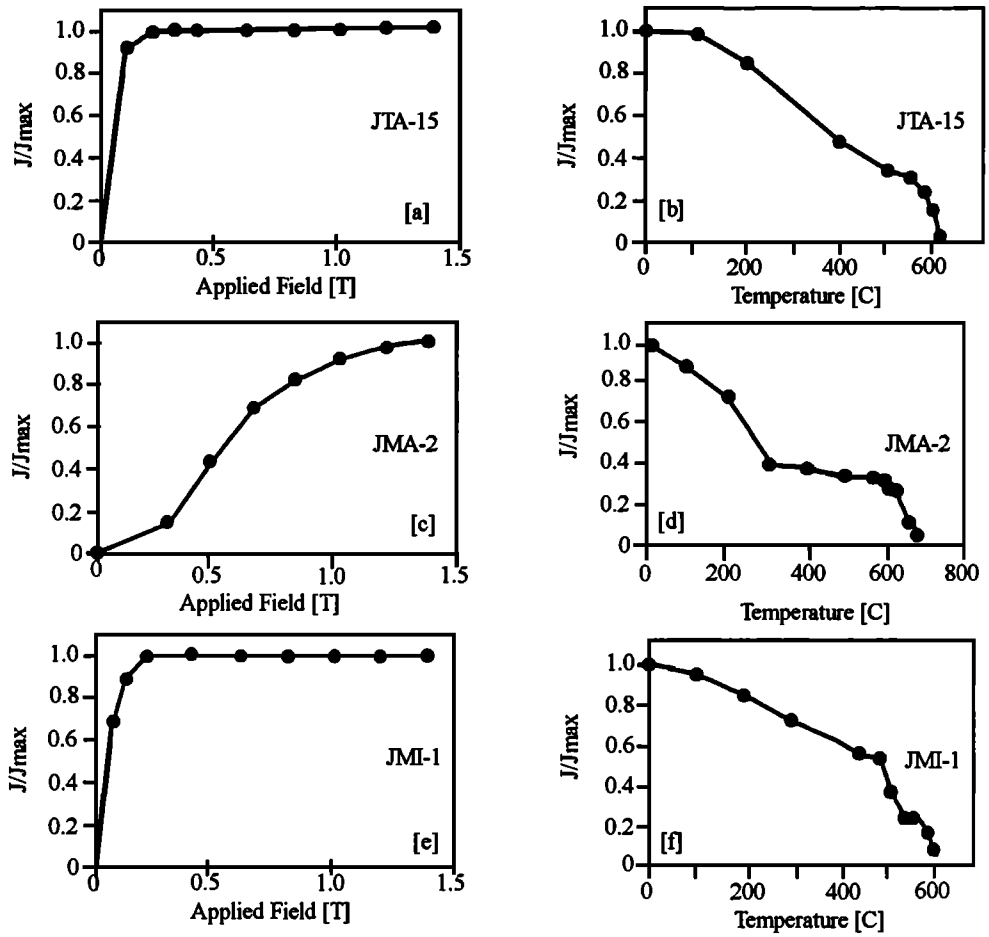


Figure 5. Isothermal remanence acquisition and unblocking temperature spectra for typical samples showing the C component (Figures 5a and 5b); the B component (Figures 5c and 5d) and the A component (Figures 5e and 5f).

DISCUSSION

Three components are observed in the Catoctin volcanics, and it is logical to infer that at least two are of secondary origin. No individual sample contained both the A and C components; however, the B component was observed in samples also containing the C component. We infer that the Catoctin C and B remanences are remagnetizations due to the Taconic and Carolinian-Penobscottian tectonic events respectively. The Catoctin C pole at 19° N, 143°E falls near the Ordovician (450 Ma) segment of the Laurentian APWP of *Van der Voo* [1990] at 16°N, 144°E (Figure 9). This age, as estimated from the APWP, corresponds to the isotopically determined Rb-Sr resetting ages of 450-470 Ma [*Badger and Sinha*, 1988; *Mose and Nagel*, 1984] and the peak of greenschist metamorphism in the Blue Ridge [*Glover*, 1989]. This pole resulted from a thermal or chemical resetting of the original remanence. The C component direction was also detected by *Brown and Van der Voo* [1982] in the Unicoi basalts of similar age in Tennessee. They concluded, as we have, that this direction is the result of a Taconic remagnetization.

The B component passes a fold test and is carried by hematite. This hematite-carried remanence was able to survive the later greenschist-grade Taconic overprint. Its direction is similar to the hematite direction found in the Unicoi basalts in Tennessee by *Brown and Van der Voo* [1982] that yielded a paleomagnetic pole at 0.4°S, 178°E. They concluded that the hematite direction found in the Unicoi was the result of high-temperature oxidation of the basalts at the time of cooling. Combining the Catoctin B component pole of 4° S, 193°E with the Unicoi basalt pole gives 2°S, 186°E. This combined mean pole falls near a Late Cambrian mean pole for Laurentia (3° N, 164° E; A95=12°). We consider that the B component is best interpreted as the result of a remagnetization during a Late Cambrian (505-515 Ma) orogenic event in the central Appalachians such as that proposed for the collisions of various arc-terranes with the Laurentian margin [*Glover*, 1989; *Rast*, 1989]. The C component pole (450 Ma) fails the fold test and the B component pole (505 Ma) passes the fold test suggesting that Taconic deformations in this region of the Blue Ridge were significant.

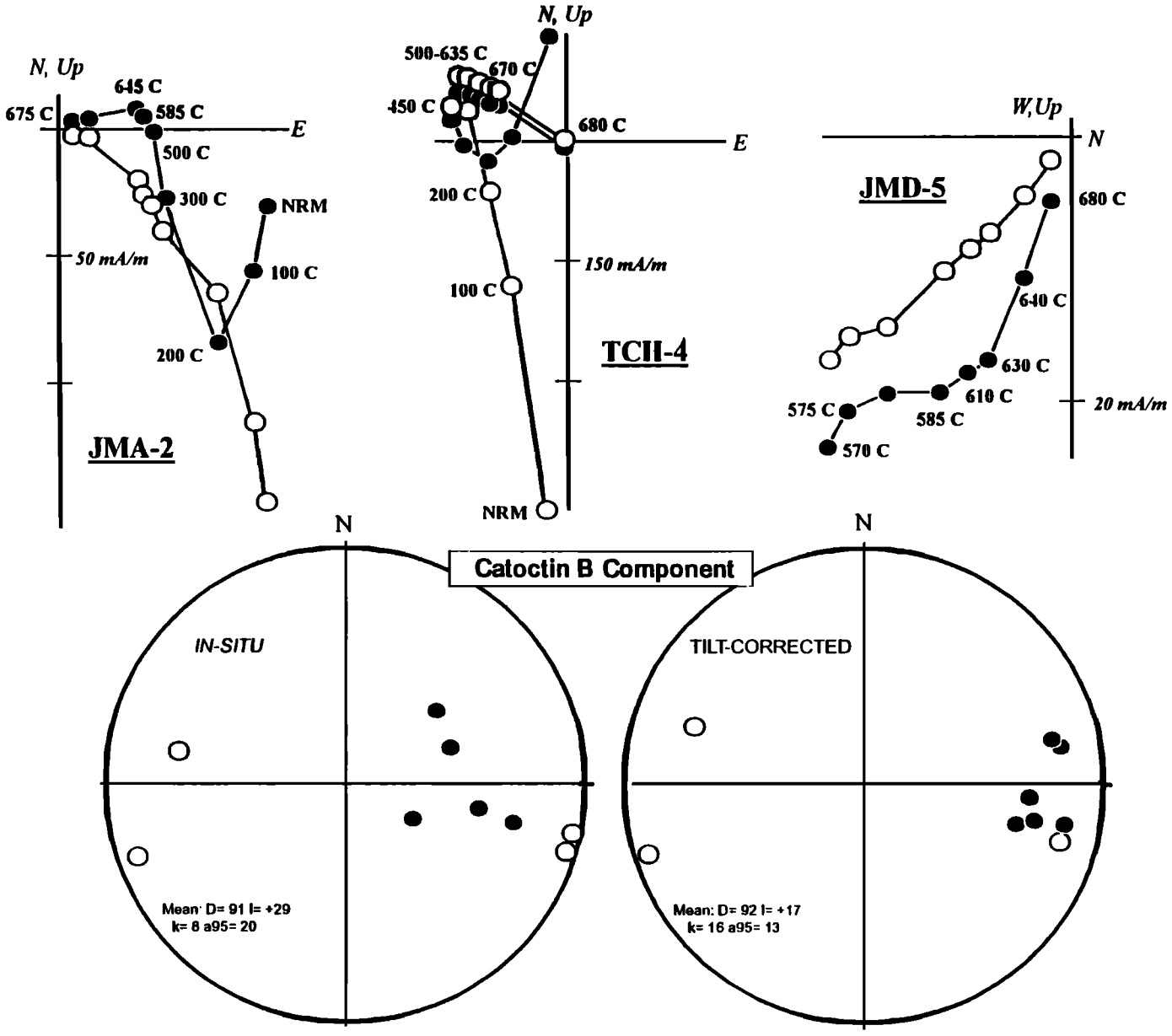


Figure 6. Typical tilt-corrected orthogonal vector diagrams of samples showing the B component of magnetization along with equal-area stereographic projections of the in situ and tilt-corrected site mean directions.

TABLE 2. Paleomagnetic Results for the A-Component

Site	n/N	Ds	Is	Dc	Ic	k	α95	Plts	Plgs	Pltc	Plgc
2	3/5	191	-81	256	-74	504	06	56 S	118 E	39 S	151 E
3	6/6	289	+52	281	+80	154	06	33 S	040 E	40 S	087 E
6	1/6	041	-79	133	-80	****	***	21 S	097 E	50 S	090 E
8	6/9	164	+89	164	+89	43	08	36 S	113 E	36 S	113 E
9	5/6	55	+80	86	+76	334	05	48 S	137 E	36 S	145 E
15	4/6	194	-76	203	-86	97	09	63 S	125 E	45 S	116 E
16	1/5	297	+82	159	+82	****	***	44 S	091 E	24 S	117 E
28	4/5	189	-75	248	-75	730	04	66 S	123 E	44 S	147 E
Mean-A	30/48	330 k=26	+83 α95=11	64 k=55	+86 α95=7			50 S	099 E	42 S	123 E
Mean-B	28/37	346 k=23	+81 α95=14	68 k=59	+84 α95=9			56 S	099 E	43 S	128 E

Notes: Mean-A is calculated using all the tabulated results for the A-Component giving each site unit weight. Mean-B is calculated without sites 6 and 16 which only contained 1 sample. The symbols used here are the same as listed in Table 1.

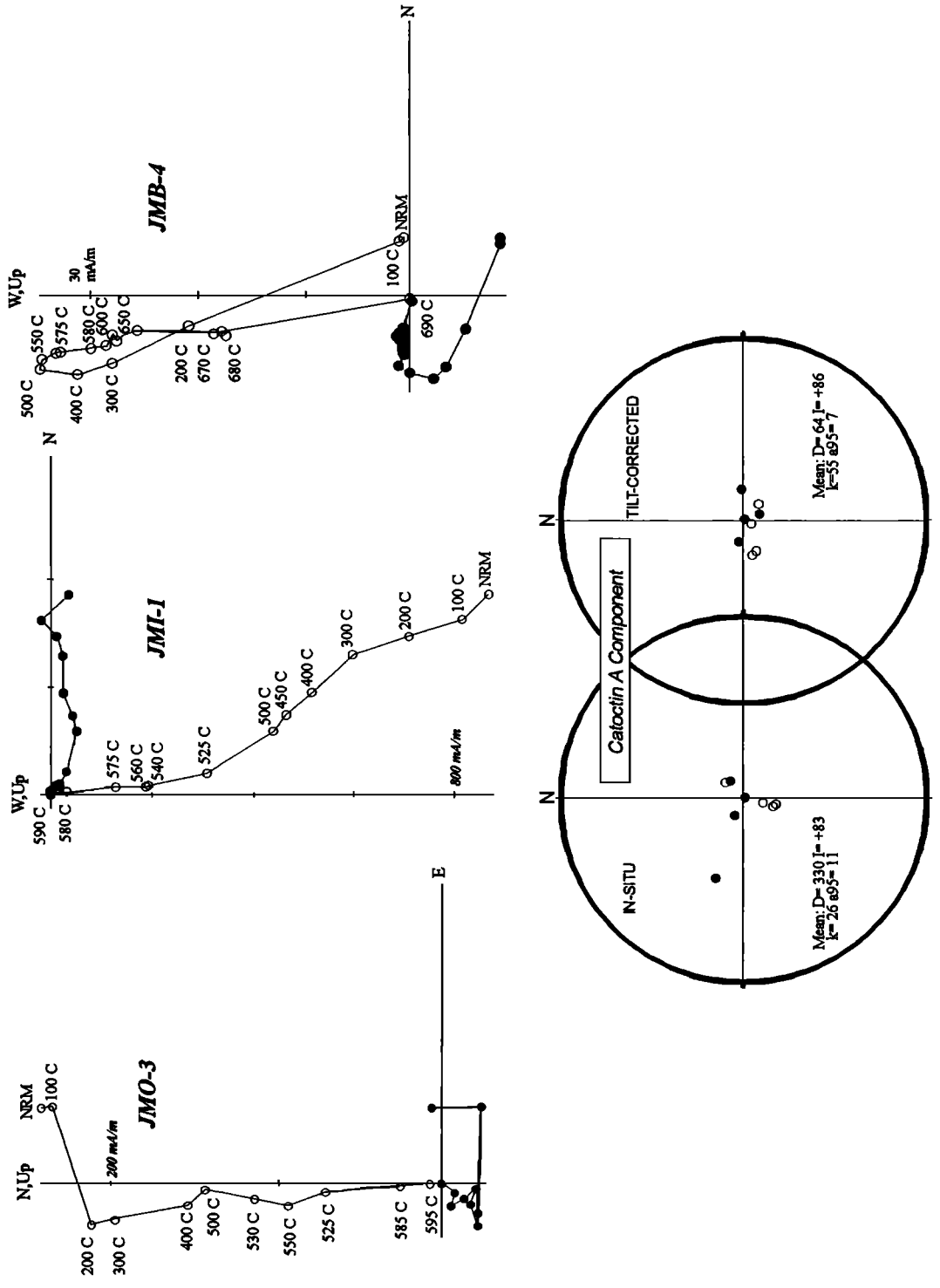


Figure 7. Typical tilt-corrected orthogonal vector diagrams of samples showing the A component of magnetization along with equal-area stereographic projections of the in situ and tilt-corrected site mean directions.

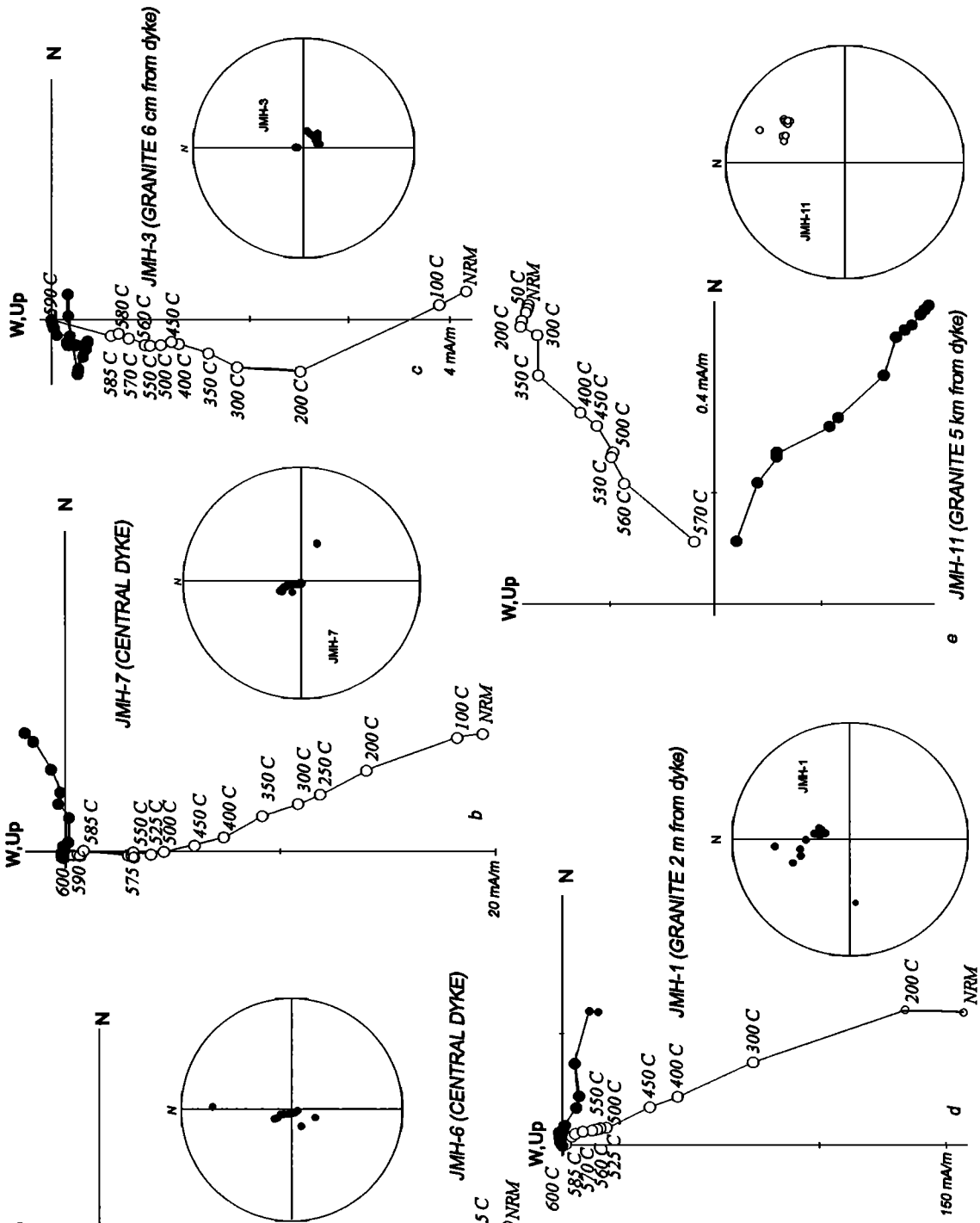


Figure 8. In situ orthogonal vector plots for samples (Site 8) collected for a baked contact test; (a) and (b) samples of dike material, (c) sample of Pedlar granite located 6 cm from the dike contact, (d) Pedlar granite sample 2 m from the dike contact and (e) Pedlar granite sample 5 km from the dike at Site 8.

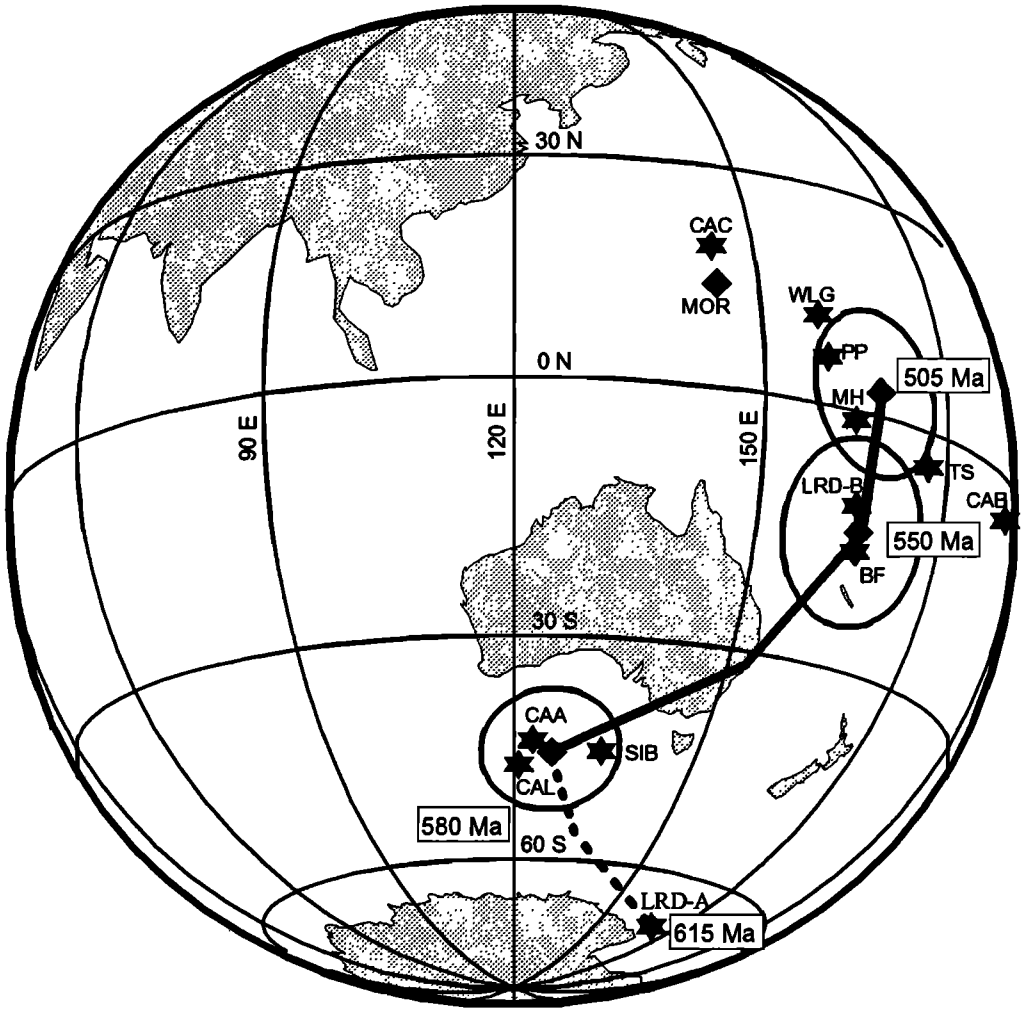


Figure 9. Apparent polar wander path for Laurentia (615-505 Ma) based on data compiled in Table 3. The solid diamond represents preferred mean poles for 580 Ma, 550 Ma, 505 Ma and 450 Ma. The key to the paleomagnetic poles is: LRD=Long Range dikes I pole (see text); CAL= Callander Complex; SIB= Sept Îles B-pole; CAA= Catoctin A-pole; BF= Buckingham flows tilt-corrected; LRD-B= Long Range dikes pole (see text); CAB= Catoctin B-pole; TS= Taum Sauk pole; MH= Moores Hollow pole; PP= Points Peak pole; WLG= Welge Creek-Wilberns pole; MOR= Middle Ordovician mean pole of *Van der Voo* [1990]; CAC= Catoctin C-pole.

The A component is interpreted to be the oldest magnetization acquired by the rocks in the Catoctin volcanic province. This component passes fold and reversals tests and may well be a primary remanence given the positive baked contact test with the older Pedlar Granite. We, therefore, consider the age of this pole at 43°S 128°E to lie between 570 and 600 Ma.

The Vendian-Early Cambrian portion of the Laurentian APWP is the subject of recent discussion [*Symons and Chiasson*, 1991; *Murthy et al.*, 1992; *Park*, 1992] due to uncertainties in both the paleomagnetic data and in placing the exact boundary between the Vendian and Early Cambrian in a radiometrically calibrated timescale. Laurentia is included in all recent models of a Late Precambrian supercontinent [*Hoffman*, 1991; *Piper*, 1987; *Dalziel*, 1991], but these models do not specifically agree on either its position relative to other continents or its paleolatitude. Results from this study were combined with previous studies (Table 3) to define the 615 to 505 Ma portion of the Laurentian APWP.

Based on their paleomagnetic investigation of the Callander Complex (Figure 3), *Symons and Chiasson* [1991] recently suggested that Laurentia was located at the south pole and thereby challenged the previously proposed equatorial Vendian position for Laurentia [e.g., *Van der Voo et al.*, 1984]. Its pole at 46°S, 121° E falls near the Catoctin A pole (Figure 9). The Callander Complex is dated at 577 Ma (U-Pb) [*Kamo et al.*, 1991] and is, therefore, similar in age to the Catoctin Formation. *Symons and Chiasson* [1991] also pointed out that the Sept-Îles B pole [*Tancyzk et al.*, 1987], when corrected for a small regional tilt, also falls near the Callander and Catoctin poles at 44°S, 134°E. The Sept-Îles Complex is not as well

TABLE 3. List of Paleomagnetic Poles for the Interval 615-505 Ma

Pole Name	Pole Latitude	Pole-Longitude	Age Range	A95	Q[a]	Paleolatitde[b]	Reference
1. Long Range I[c]	69° S	170° E	613-617 Ma	15°	3	41° S	*Murthy et al.*, 1992
2. Catoctin A	43° S	128° E	573-597 Ma	9°	6		this study
3. Callander Complex	46° S	121° E	577 Ma	3°	6		*Symons & Chiasson*, 1991
4. Sept-Îles B[d]	44° S	134° E	c. 580 Ma	10°	3		*Tanczyk et al.*, 1987
Mean 2-4	**45.5° S**	**126° E**	**580 Ma**	**8°**	--	**58° S**	
5. Buckingham Flows[e]	16° S	165° E	c. 550 Ma	8°	4		*Dankers & LaPointe*, 1981
6. Long Range Dikes	11° S	164° E	c. 550 Ma	18°	3		*Murthy et al.*, 1992
Mean 5-6	**13.5° S**	**164.5° E**	**550 Ma**	--	--	**14° S**	
7. Welge-Wilberns	11° N	158° E	505-523 Ma	10°	5		*Loucks & Elmore*, 1986
8. Point Peak-Wilberns	6° N	159° E	505-523 Ma	5°	5		*Van der Voo et al.*, 1976
9. Taum Sauk Ls.	4° S	176° E	505-523	7°	5		*Dunn & Elmore*, 1985
10. Moores Hollow	1° S	163° E	505-523	9°	5		*Farr & Gose*, 1991
Mean 7-10	**3° N**	**164° E**	**505 Ma**	**12°**		**6° S**	

Notes:
[a]Q-Value [quality factor] assigned according to Van der Voo [1990]
[b]Paleolatitudes calculated for a central point on Laurentia [45° N, 260° E]
[c]Long Range Dikes pole calculated from dike dated at 615 Ma; see text for complete details
[d]Sept-Îles B-pole recalculated according to Symons and Chiasson [1991]
[e]Buckingham flows recalculated; see text for details

dated as the Callander Complex and Catoctin Formation. Its suggested age is about 580 Ma [*Higgins and Doig*, 1981]. Further support for a new 580-600 Ma pole for Laurentia comes from unpublished results from the 570 Ma Manitou Islands Complex (Ontario, Canada). A preliminary paleomagnetic pole for this unit falls at 40°S, 115°E [D.T.A Symons, personal communication, 1993]. Our new Vendian pole (570-600 Ma) for Laurentia is computed as the average pole from the Callander, Sept-Îles B, and Catoctin A poles at 45.5°S, 126°E (A95=8°) and places the Callander Complex at 72° ± 8°S, the Catoctin Formation at 72° ± 8°S and the Sept-Îles intrusion at 81° ± 8°S.

In a recent study of the Long Range dikes, *Murthy et al.* [1992] reported "anomalous" results from three sites in one of the dikes. This anomalous result was derived from a dike that was dated at 615 ± 2 Ma (U-Pb) [*Kamo et al.*, 1989]. Although it is possible that secular variation was not averaged in the results from these three sites, the pole at 69°S, 170°E appears to give a reasonable approximation of the 615 Ma pole for Laurentia (Figure 9).

The remaining portion of the Laurentian APWP is represented by the Buckingham flows in Quebec and the Long Range dikes "characteristic" pole. The Buckingham flows are imprecisely dated at 572 ± 32 Ma (whole rock K-Ar) [*Lafleur and Hogarth*, 1981] and the in situ paleopoles were originally reported as three separate poles by *Dankers and Lapointe* [1981]. A tilt correction of these data was not performed because the dip of the flows was considered primary. However, tilt correction of all sites using the bedding data in *Dankers and Lapointe* [1981] gives a mean pole for the Buckingham volcanics at 16°S, 165°E. Results from *Murthy et al.* [1992] taken from one of the dikes in the Long Range series with a K-Ar biotite age of 553 ± 22 Ma [*Wanless et al.*, 1970] gives a paleomagnetic pole at 11°S, 164°E, nearly identical to the Buckingham pole. We consider these K-Ar ages to be representative of an Vendian/Early Cambrian age of 550 Ma; moreover, their mean pole at 13.5°S, 164.5°E indicates an intermediate position of Laurentia between its high-latitude position at 580 Ma and the 505 Ma equatorial position. The paleolatitude for a central point on Laurentia at 550 Ma is approximately 15°S [Table 3]. Additional support for this segment of the path is provided by the Johnnie Formation (Rainstorm Member) of Nevada studied by *Gillett and Van Alstine* [1982]. The paleomagnetic pole (unrotated) for the Johnnie-Rainstorm is 10°S, 162°E. The age of the Rainstorm Member is not well known; however, if the Vendian/Cambrian boundary is moved forward to 540 Ma as suggested by *Conway Morris* [1989] then the age of the Rainstorm Member may be about 550 Ma [*Link et al.*, 1992]. Because the age of the Vendian/Cambrian boundary is still the subject of debate and there is uncertainty about the age of the Rainstorm Memeber, this result was not used in calculating our mean 550 Ma pole.

The 505 Ma mean pole for Laurentia was calculated from results compiled by *Van der Voo* [1990] with Q≥3 and the Moores Hollow result of *Farr and Gose* [1991]. Paleomagnetic results that require rotation such as the Florida

Mountains pole of *Geissman et al.* [1991] and the Colorado Intrusives pole of *French et al.* [1977] were not used although these poles fall near our 505 Ma mean pole. The Catoctin B pole falls near the Late Cambrian [505 Ma] mean pole for Laurentia at 3°N, 164°E. This Late Cambrian mean pole places Laurentia near the equator (Table 3).

Our proposed APWP (Figure 9) is interpreted to indicate that the Vendian rifting event along the eastern margin of Laurentia took place at high southerly latitudes. The "breakout" of Laurentia included rapid drift of the continent toward the equator. The minimum rate of drift with respect to the pole calculated for this motion is 16 cm yr^{-1} during the interval 600-550 Ma. This drift rate is high compared to modern day drift rates, but not unparalleled in the Phanerozoic [*Klootwijk et al.*, 1992; *Meert et al.*, 1993; *Chen et al.*, 1993]. The Varangian-Ice Brook glaciations on Laurentia were previously considered as having occurred at low latitudes. Current estimates for the timing of these events range between 650 and 610 Ma [*Link et al.*, 1992]. Our new apparent polar wander path suggests that these glaciations may have taken place at much higher paleolatitudes than previously thought.

The next older well-dated poles for Laurentia are from the 725 ± 5 Ma Franklin

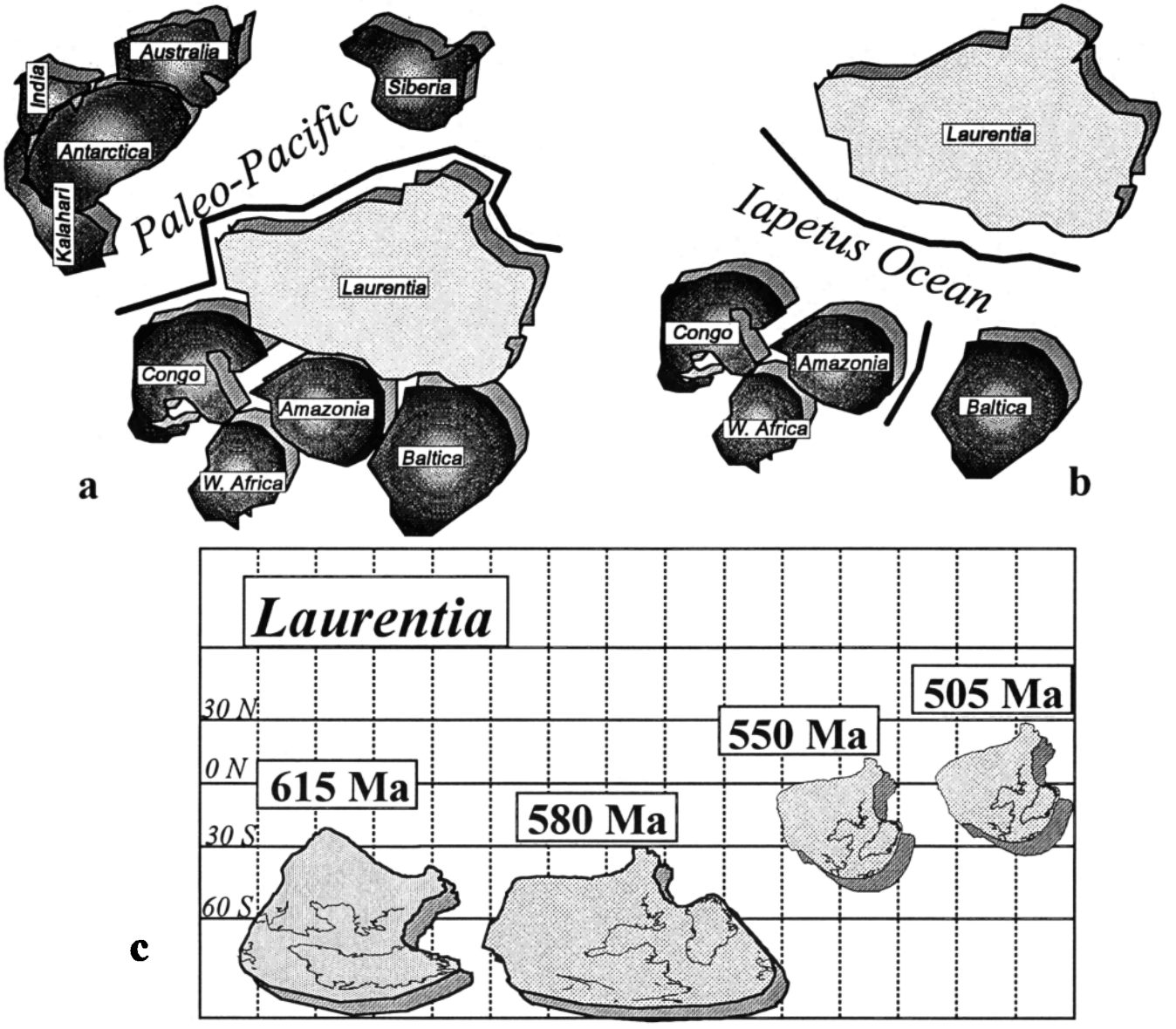


Figure 10. (a) Laurentia at ca. 650 Ma following the hypothesized rifting of proto-East Gondwana away from western Laurentia and prior to the second phase of rifting along the present-day eastern margin; (b) Laurentia following the second rifting phase with passive margin development along the present-day east coast at ca. 550 Ma and; (c) Drift history of Laurentia from 615 Ma to 505 Ma indicating a period of rapid drift away from the south pole between 580 and 550 Ma.

diabases [*Fahrig et al.*, 1971] and Natkusiak Formation [*Palmer et al.*, 1983]. There is, therefore, a gap of 110 m.y. between well-dated paleomagnetic poles for Laurentia. This gap makes it difficult to link our path with the earlier Proterozoic path. A recent analysis of the 725-830 Ma portion of the Laurentian APWP is provided by *Powell et al.* [1993]. All of the mean poles calculated in that study show evidence for primary magnetization despite their resemblance to the Late Cambrian pole reported herein. *Park* [1992] debated the similarity of the 725-830 Ma poles to the 505 Ma poles for Laurentia by proposing an extensive track for the Laurentian APWP across the south pole rather than invoking a later, Cambrian, remagnetization. However, as pointed out by *Powell et al.* [1993], this interpretation requires either a reconfiguration of Rodinia or a reversal of the polarity of all pre-Devonian Gondwana paleomagnetic poles. At present it is difficult to link our path to the earlier portion of the Laurentian APWP without more data from the interval 615-725 Ma.

Bond et al. [1984] proposed the inclusion of Laurentia as part of an Vendian supercontinent, centered at the south pole. Subsequent reconstructions placed Laurentia at the equator [*Dalziel*, 1991] or the south pole [*Dalziel*, 1992]. The available paleomagnetic data support the south polar reconstruction of *Dalziel* [1992] for Laurentia with the 600 Ma rifting event responsible for the breakout of Laurentia from western Gondwana. Figure 10 shows the drift and tectonic history for Laurentia based, in part, on the paleomagnetic data given in Table 3. Figure 10a shows Laurentia at ca. 650 Ma following the initial rifting of proto-East Gondwana away from the present-day western margin to split the supercontinent shown in Figure 2. Figure 10b shows Laurentia following the second rifting phase with a passive margin along the present day east coast developing around 550 Ma. Figure 10c shows the drift history of Laurentia from 615 to 505 Ma, based on paleomagnetic data, demonstrating a period of rapid drift away from the south pole between 580 and 550 Ma.

CONCLUSIONS

The rocks of the Catoctin volcanic province of Virginia yield three paleomagnetic poles that are interpreted to correlate with a sequence of tectonic events at 570-600 Ma, 505-510 Ma, and 450 Ma. The primary 570-600 Catoctin A pole amplifies the Vendian APWP for Laurentia suggested by *Symons and Chiasson* [1991] and *Park* [1992]. These new data indicate that Laurentia was located at the south pole at 615 Ma and drifted rapidly toward the equator during the Vendian to Early Cambrian interval (580 to 550 Ma). Plate velocities calculated for Laurentia during this interval are at least 16 cm yr^{-1} suggesting that large continents may travel at fast rates for short periods of time. This high-latitude positioning of Laurentia during the Vendian suggests that the Ice Brook-Varangian glaciations do not belong to the group of anomalously low-latitude glaciations proposed for other continents during this interval [*Harland*, 1983].

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